Interpreting Clinically Significant Changes in Patient-Reported Outcomes

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BACKGROUND. The goal of this study was to determine what magnitude of change in a patient-reported outcome score is clinically meaningful, so a clinicians' guide may be provided for estimating the minimal important difference (MID) when empiric estimates are not available.

METHODS. Consecutive laryngeal cancer patients (n = 98) rated their quality of life (QOL) relative to other patients. These comparisons were contrasted with arithmetic differences in scores on the Functional Assessment of Cancer Therapy-Head and Neck (FACT-H&N) scale, Functional Assessment of Cancer Therapy-General (FACT-G) scale, 2 utility measures (the time tradeoff [TTO] and Daily Active Time Exchange [DATE]), and performance status (Karnofsky) scores.

RESULTS. The FACT-H&N score needed to differ by 4% for average patients to rate themselves as "a little bit better" relative to other patients (95% CI, 1%–8%) and by 9% to rate themselves as "a little bit worse" relative to others (95% CI, 4%–13%). The corresponding values for other measures were FACT-G 4% (1%–7%) and 8% (95% CI, 5%–11%); TTO 5% (95% CI, 0%–11%) and 6% (95% CI, 0%–10%); DATE 5% (95%CI, 2%–9%) and 14% (95% CI, 0%–5%); Karnofsky 4% (95% CI, 1%–6%) and 10% (95% CI, 7%–13%). In each case, the minimal important difference (MID) was about 5% to 10% of the instrument range.

CONCLUSIONS. One rule of thumb for interpreting a difference in QOL scores is a benchmark of about 10% of the instrument range. Patients appear to be more sensitive to favorable differences, so an improvement of 5% may be meaningful. This simple benchmark may be useful as a rough guide to meaningful change. ***Cancer* 2007;110:196–202.** © *2007 American Cancer Society.*

KEYWORDS: quality of life, health utility, outcomes. patient reported outcomes, minimal important difference, meaningful change, laryngeal neoplasms, questionnaires, head and neck neoplasms.

Patient-reported outcomes such as quality of life (QOL) and utility have been reported with increasing frequency in oncology trials. These outcomes are intended to assess the "goodness of life" from the patients' perspective. Although many clinicians agree in principle that such measures provide added value to survival data, it remains difficult to define what magnitude of change in such data is clinically important.[1]

Minimal important difference (MID) refers to the smallest difference that reflects a clinically important change in a score.[2] Having some knowledge of the MID is important for determining the sample size when planning trials with QOL endpoints and for interpreting the results of such studies when completed. In addition, whether a decision analysis on patient survival should be quality-weighted may depend on how large a difference in utility is of importance to patients.

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The MID may be estimated statistically or by comparison to accepted standards. Other approaches involve using anchor-based methods (eg, by asking a patient to compare his current status to that of another individual or to his status at a previous point in time).[3] We favor such introspective approaches because they directly reflect the point of view of the patient.[4] However, such approaches are effort intensive and their results, although of value to researchers, may not be read or easily retained by a practicing clinician.

The primary objective of this project was to investigate whether MID estimates are similar within a single group of patients for different measures of well-being. As a secondary goal, we reviewed the literature to try to devise a general rule of thumb for MID, an approximation that would be useful to the average clinician and that could be applied across QOL measures in various groups of patients.

MATERIALS AND METHODS

This project was conducted at the Princess Margaret Hospital, a quaternary-care comprehensive cancer hospital in Toronto, Canada. Details of the participants and accrual process have been previously published.[5] In brief, laryngeal cancer survivors attending for follow-up between May and November of 1998 were eligible for the study if they had squamous cell cancer stage T1-T4N0-3M0.[6]

Individuals were required to have completed curative treatment for the disease at least 6 months before study enrollment and to be free of active disease when assessed. At the initial interview, patients were excluded if they were unable to speak and read English sufficiently to complete the instruments or to converse clearly about their disease.

At the initial clinic visit, baseline Karnofsky performance status was recorded.[7] After instruction, participants completed the Functional Assessment of Cancer Therapy-Head and Neck (FACT-H&N) questionnaire.[8] The time trade off (TTO), a widely accepted utility measure, was also completed.[9–11] A second, experimental, utility measure was used, which we have termed the Daily Active Time Exchange (DATE).[12,13]

After initial assessment, each patient was offered a choice of dates for a group meeting. We made no conscious effort to influence the composition of patient groups or the order of patient interactions. At the group meeting, patients were instructed to serially speak one-on-one with each group member and to discuss any issues they felt were important to their health or QOL. They were asked to share descriptions of their current well-being and any problems they may be having as a result of their disease or treatment.

Each patient was serially paired with every other patient in the group for interactions that lasted about 10 minutes. After each pairing, patients separated and confidentially rated themselves compared with the person with whom they had just met (patients were encouraged not to share their ratings). Rating forms were collected after each round, after which patients were paired with different partners and the process was repeated. The ratings were made on a Likert scale with values from 1 (much better) to 7 (much worse). Value 4 represented "about the same,"; Value 3 and Value 5 represented "a little bit" better and worse, respectively; and Value 2 and Value 6 were defined as "somewhat" better and worse, respectively. This scale has been used previously for comparisons both between individuals and over time in a single individual.[4,14–20]

Sample size was based on previous studies that used a similar methodology and included 5 groups of 20 patients each (for a total of 100 patients). For each paired comparison, the arithmetic difference between the 2 individual scores was calculated for FACT-H&N, Functional Assessment of Cancer Therapy-General measurement (FACT-G), and subscales of FACT-G, TTO, DATE, and Karnofsky score. Each pairing generated 2 contrasts, that is, A's rating compared with B and B's rating compared with A. The unit of analysis for MID quantifications was the contrast rather than the patient.

As individual data points were expected to be somewhat scattered, we calculated mean arithmetic differences for each level of subjective comparison rating. For example, for patients who rated themselves as "about the same" (4, on a 7-point scale), we calculated the group mean for each arithmetic difference variable. We undertook the same procedures for each level (1 to 7) of comparison rating.

We were then able to calculate an estimate of MID. For the TTO, for example, the mean arithmetic difference in scores for a comparison rating of 4 ("about the same") was subtracted from the mean arithmetic difference in scores for a comparison rating of 3 ("a little bit better") to obtain an estimate of MID for the TTO. A second estimate of the MID was calculated by a difference of means for patients rating themselves as "a little bit worse" compared with "about the same." Student's *t*-test was used to compare these 2 "positive" and "negative" estimates of MID. By using the same process described above, we estimated the MID for the DATE, FACT-H&N, Karnofsky score, FACT-G, and subscales of FACT-G (subscale data not shown).

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TABLE 1
Estimates of Positive and Negative MID*

Instrument	Range	Positive MID (95% CI)	Negative MID (95% CI)	*P*
FACT-H&N	0–144	6.22 (1.42–11.02)	12.40 (5.09–19.71)	.02
FACT-G	0–108	4.37 (1.00–7.74)	8.28 (5.45–11.65)	<.0001
TTO	0–1	0.050 (0–0.111)	0.060 (0–0.127)	<.0001
DATE	0–1	0.055 (0.018–0.092)	0.140 (0–0.542)	<.0001
Karnofsky	0–100	3.83 (1.16–6.50)	10.35 (7.53–13.16)	<.0001

MID indicates minimal important difference; FACT-H&N, Functional Assessment of Cancer Therapy-Head and Neck scale; FACT-G, Functional Assessment of Cancer Therapy-General measurement scale; TTO, time trade off; DATE, Daily Active Time Exchange; Karnofsky, Karnofsky performance status scale.

* Positive MID indicates the smallest favorable difference on each instrument that was noticeable to the average patient, and negative MID indicates the smallest unfavorable difference on each instrument that was noticeable to the average patient.

TABLE 2
Positive and Negative MID as a Percentage of Total Instrument Score*

Instrument	Positive MID	Negative MID
FACT-H&N	4.3%	8.6%
FACT-G	4.0%	7.7%
TTO	5.0%	6.0%
DATE	5.5%	14%
Karnofsky	3.8%	10%

MID indicates minimal important difference; FACT-H&N, Functional Assessment of Cancer Therapy-Head and Neck scale; FACT-G, Functional Assessment of Cancer Therapy-General measurement scale; TTO, time trade off; DATE, Daily Active Time Exchange; Karnofsky, Karnofsky performance status scale.

* Absolute values of MID are indicated as a percentage of maximum possible score on each instrument, eg, for FACT-H&N, positive MID is 6.22 ÷ 144 = 4.3%.

To assess the degree to which our estimates of MID derived from the whole group of participants could be applied to individual patients, we plotted the cumulative probability of rating oneself as at least "a little bit better" for several levels of arithmetic difference in FACT-H&N, TTO, DATE, and Karnofsky scores.

Statistical analysis was carried out using Statview software (SAS Institute, Cary, NC) on a Macintosh Performa 580 computer (Apple, Cupertino, Calif). All data was doubly entered, and inconsistencies were reconciled by checking original hard copies. Approval to conduct this study was granted by both the Princess Margaret Hospital Clinical Trials Committee and the University of Toronto Oncology Human Review Committee. All participants gave informed consent.

RESULTS

Ten group meetings took place, 3 with 8 participants, 3 with 9 participants, and 1 with each of 7, 10, 14, and 16 participants, respectively. The maximal possible number of contrasts was 938. After consideration of missing data, 861 of 938 (92%) of contrasts were analyzable for FACT-H&N, 823 of 938 (88%) for TTO, 847 of 938 (90%) for DATE, and 861 of 938 (92%) for Karnofsky. Post-hoc power calculation suggests that this provided adequate power to detect correlations (between arithmetic differences in score and subjective comparison ratings) of 10% or higher, which explain about 1% or more of total variance.[21]

Similar to past research, the distribution of subjective comparison ratings was highly skewed. The majority of individuals rated themselves as "about the same" as their partners. More often, individuals rated themselves as at least "a little bit better" rather than "a little bit worse" than their partner. Figure 1 shows a statistically significant correlation ($r = -0.22$) between differences in TTO score and subjective comparison ratings. Similar results were found for other instruments (data not shown).

Table 1 shows our estimates of positive and negative MID for the outcomes measured in this study, expressed as absolute values, and shows the range of scores for each outcome. In Table 2, the MID values have been converted into percentages of total score for each instrument. With the single exception of 1 subscale (emotional well-being) of FACT-H&N, for each outcome the negative MID was significantly larger than the positive MID.

We plotted the cumulative probability of rating oneself as at least "a little bit better" on the subjective comparison of overall well-being versus the difference in score on FACT-H&N, TTO, and DATE. Figure 2 shows the resulting curve for TTO. This shows that at the estimated positive MID of 0.05, just over half of our participants rated themselves as at least a little bit better than their partners. However, if the difference was at least 0.15, about 75% of the ratings were at least "a little bit better." Similar curves were found for FACT-H&N and DATE. Therefore, some patients are more sensitive to a difference in QOL or utility score than are others, thus limiting the applicability of average estimates of MID to individuals.

DISCUSSION

Patient-reported outcomes provide important, yet ambiguous, information to guide treatment decisions. What threshold should be used to claim that a new treatment is associated with better QOL than an established one? In this study, we assessed the threshold at which a difference in QOL or utility scores was associated with a small, but noticeable, difference in perceived well-being for laryngeal can-

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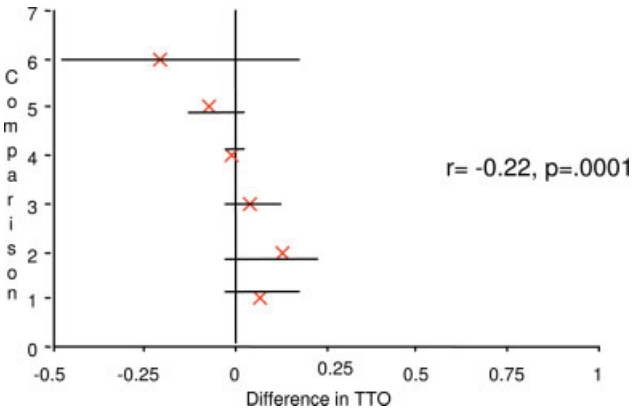


FIGURE 1. These are mean differences in time trade off (TTO) scores with 95% confidence intervals for each subjective comparison. Subjective comparison definitions are 7, "much worse,"; 6, "somewhat worse,"; 5, "a little bit worse,"; 4, "about the same,"; 3, "a little bit better,"; 2, "somewhat better"; 1, "much better.".

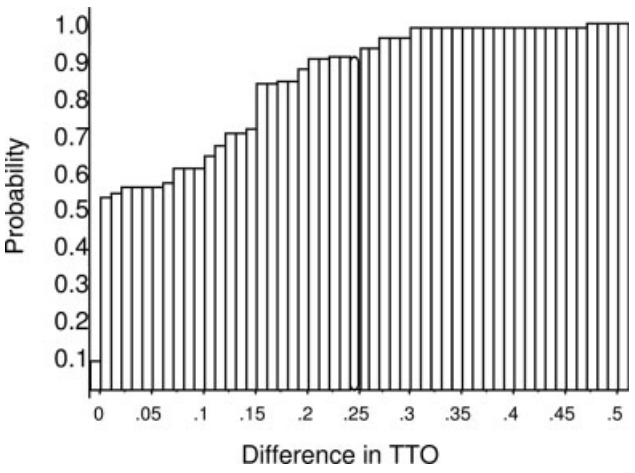


FIGURE 2. Cumulative probability of a patient rating himself at least "a little bit better" is illustrated for a given arithmetic difference in time trade off (TTO).

cer patients. By using an interpatient method (ie, comparisons between individuals at a single time point) and by comparing arithmetic differences in QOL with subjective comparison ratings, we found thresholds ranging from 4% to 14% of the possible span of each instrument.

For the 36-item FACT-H&N, with its minimal score of 0, maximal score of 144, and 5-point Likert scale, the positive and negative MID estimates of 6 (4%) and 12 (9%) correspond to a mean change of about 0.2 to 0.3 per question or 1 point on every third to fifth question.

For the TTO, the MID estimate of about 0.05 (5%) corresponds to giving up 1 year from 20 remaining life years. For DATE, and assuming 8 hours of usual sleep daily, the positive MID of 0.05 (5%) and negative MID of 0.14 (14%) represent about 48 minutes and 134 minutes of additional daily sleep requirements.

A previous study of the MID in the FACT-G used an anchor-based method, assessing change within patients over time.[22] Despite a difference in methodology, the MID was found to be similar to our estimates, showing a positive MID of 5.48 (4.9%) and a negative MID of 9.87 (8.8%). Other studies have used an approach combining statistical methods and anchoring to performance status and pain anchors: In breast cancer patients, the MID for FACT-G was estimated as 5 to 6 points (5%–6%),[23] and in lung cancer patients the MID for a Trial Outcome Index constructed from the FACT-L (L, lung) was 4 (4.8%)[24]; both studies found that MID estimates for all instruments assessed were generally between 5% and 8% of instrument range.

Previous estimates of MID using the interpatient method have been made for other health questionnaires. For the 20-item Chronic Respiratory Questionnaire (CRQ) in chronic obstructive pulmonary disease (COPD) patients, MID was about 8%.[14] For the 8-item Stanford Health Assessment Questionnaire (HAQ) functional status instrument in arthritis patients, MID was about 6% in 1 study, whereas another found about a 7% positive and a 16% negative MID.[18,19] Other interpatient studies have determined MID for the 6-minute walk and the forced expiratory volume in 1 second (FEV-1), measures that do not have an absolute maximum for scoring. For reference, MID for the 6-minute walk was 40 meters (m) and a typical normal value is about 700 m, giving a MID of about 6%.[17] Similarly, for the FEV-1, the MID was 112 mL or 6% of a typically normal 2000 mL.[20]

Two studies that used the intrapatient method (ie, measuring differences within an individual over time by use of a patient-reported change score) have found MID estimates of about 7%.[4,15] For cancer patients, a percentage change of about 5% to 10% corresponded with the MID for different subscales of the EORTC-QLQ-C30, a questionnaire developed by the European Organisation for Research and Treatment of Cancer's Group for Research in Quality of Life.[16]

A recent review has also noted that most published MID estimates fall into the range of 5% to 10% of instrument range.[25] Considering differences in patients, disease, and instruments in which these MID studies have been done, the similarity of percentage change estimates of MID is remarkable. Our

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estimates for the FACT-H&N, the Karnofsky performance status scale, and 2 utility instruments are well within the previously reported percentage range for other QOL instruments.

We are not aware of any publications that have attempted to assess the MID for the TTO or DATE. However, 1 study did establish MID estimates for the multiattribute utility measures SF-6D (range, 0.29–1.00 [Short-Form 6-Dimensions; Sheffield Health Economics Group, School of Health and Related Research, The University of Sheffield, Sheffield, UK]) and EQ-5D (range, −0.59–1.00 [EuroQol Business Management, Rotterdam, Netherlands]). Despite differences in these 2 instruments and their ranges, the MID values were found to be proportionally equivalent at 5.8% and 4.6% of the instrument range, respectively.[26] Our estimates of 5% to 6% for the TTO and 5% to 14% for the DATE are similar.

In 1 arthritis trial, a larger "negative" than "positive" MID was found.[19] Our study consistently showed that a deterioration must be larger than the corresponding improvement for it to be detectable to the average patient. This held true for both site-specific (FACT-H&N) and general QOL measures (FACT-G), for both utility measures (TTO and DATE), and for the Karnofsky performance status scale. Other studies have shown similar trends but not always with statistical significance.[14,17,18,20,22,26]

In most studies that use the intrapatient method, including the single study with this methodology involving cancer patients, positive and negative changes were not reported separately.[4,16] In asthma patients, positive and negative MID were reported for 4 domains, and no consistent trend was observed between their magnitudes.[15] However, most studies that use the intrapatient method are poorly designed to test for possible asymmetries of the MID. We wonder whether a small improvement in QOL may indeed be more meaningful to cancer patients than a small deterioration. Improvements are welcomed and encouraging, whereas deterioration is disappointing and protective adaptation may take place. Such a difference in value would be consistent with ego bias.[27]

It is possible that the trend to a higher MID for negative differences may be due to the skewed distribution of responses to subjective comparisons.[27] Patients tended to rate themselves as "better" rather than "worse" compared with others, so that fewer estimates of negative MID were possible. However, the contrast we observed between negative and positive MID is unlikely to be due to the small sample size. An alternative explanation for the larger negative MID values may be that patients who did rate themselves as worse were anomalous individuals with a negative self-impression, which led them to underestimate their QOL on the self-rated measures. However, this conjecture is implausible given the consistency of our findings on different measures and in all patient subgroups.

We note that the emotional well-being subscale of FACT-H&N, unlike all other metrics assessed, did not exhibit a difference in positive and negative MID. This subscale also showed the lowest correlation with its corresponding subjective comparison rating, and we hypothesize that the brief encounters used in our study provided a better opportunity for participants to judge physical and functional issues, rather than internal emotions.

We hypothesized that the MID, expressed as a percentage of the maximal instrument range, may be smaller for a multi-item index as opposed to a single-item measure. Our data did not show such a relation, with all of our outcome measures showing MID estimates in the 4% to 6% range for positive MID and the 6% to 14% range for negative MID. Previous estimates of MID have fallen within a similar range, thus suggesting that patients may be able to detect a constant proportional change on different measures. This contrasts with statistical methods of estimating MID, which may give smaller estimates for multi-item scales because of their higher reliability and smaller standard deviations.[1]

Other methods for estimating MID have been proposed. Cohen[28] has suggested that the lowest theoretical bound of the MID is about 20% of the standard deviation (SD), and based on his work, Sloan and colleagues have recommended 33% SD as a more realistic lower bound for a clinically important change.[29] Norman suggests 0.5 SD.[30] Wyrwich suggests the use of 1 standard error of measurement (SM) as the MID.[31] Table 3 shows calculations of estimates of MID using these suggested methods for our data. In general, the differences that were noticeable to our patients were larger than statistical methods would suggest, but smaller than the 0.5 per item rule suggested by Jaeschke and Guyatt.[4] The simple method of estimating MID as 5% to 10% of the instrument range performed at least as well as the other suggested methods.

As previously noted, estimates of MID do not apply to all individuals.[20] To be uniformly detectable by nearly all patients, much larger changes need to be present. However, because estimates of MID are useful mainly in planning and interpreting studies that involve groups of patients, individual diversity does not mitigate the scientific quest for group tendencies.

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TABLE 3
Comparison of Methods for Estimating MID

Instrument	33% SD	50% SD	Method of estimating MID: 0.5 Points per item	1/10 Observed range	5%–10% Maximal score	1 SM† [5]	Observed MID*
FACT-H&N	5.05	7.65	18	7.7	7.2–14.4	5.51[12]	9.31
FACT-G	3.75	5.67	13.5	6.1	5.4–10.8	3.20[8]	6.32
TTO	0.051	0.077	N/A	0.075	0.05–0.1	0.075[32]	0.055
DATE	0.040	0.060	N/A	0.053	0.05–0.1	Reliability coefficient unknown	0.097
Karnofsky	2.77	4.2	N/A	3	5–10	2.79[12,13]	7.09

MID indicates minimal important difference; FACT-H&N, Functional Assessment of Cancer Therapy-Head and Neck scale; FACT-G, Functional Assessment of Cancer Therapy-General measurement scale; TTO, time trade off; DATE, Daily Active Time Exchange; Karnofsky, Karnofsky performance status scale.

Standard error of measurement (SM) was calculated using the SD data from this patient sample[5] and published reliability coefficients for the QOL and utility instruments. For the FACT-G, test-retest reliability coefficient is 0.92 as reported in the instrument manual.[8] For the FACT-H&N, test-retest reliability coefficient was 0.82 in this sample as previously published.[12] For the TTO, internal reliability was reported as 0.77 by Torrance.[32] No published reliability coefficient was found for the DATE instrument. For the Karnofsky performance status scale, the standard deviation from this sample was 8.42,[12] and inter-rater reliability was reported as 0.89 by Schag et al.[33]

* Mean of positive and negative MID.

† SM = SD × the square root of (1 − reliability coefficient).

In contrast to statistical methods that require complex calculation and to individualized anchor-based assessments, the use of a constant, proportional estimate of MID has the advantage of providing an easily retained and clinically comprehensible benchmark. This approach, which is intended only as a rough rule of thumb useful for clinicians and consumers of QOL data, has the advantage of requiring no knowledge of statistics and can easily be applied as long as the maximal and minimal scores of a given instrument are known.

Conclusion

We conclude that the positive MID for these instruments in this setting is about 5% of the maximal instrument score, whereas the negative MID is significantly higher, at about 10% of the maximal instrument score. In our sample, a small improvement in well-being was relatively easy for patients to detect, whereas a larger deterioration needed to be present before it was noticeable.

Comparison with previous studies suggests that across patient diagnoses, the magnitude of MID, expressed as a percentage of maximal instrument score, falls within a narrow range for a variety of QOL and functional status instruments. It may be possible for clinicians to use this magnitude as a rule of thumb to interpret the clinical importance of differences in patient-reported outcomes of clinical trials in chronic diseases such as cancer. The outstanding degree of convergence observed between different measures within our study, and between our results and those obtained by others using very different methods, suggest that a high degree of confidence can be used in applying these guidelines to the interpretation of patient-reported outcome data. Confirmation of this range in further trials will be necessary, however, before this range can be used to estimate MID for novel instruments without the necessity of carrying out an explicit MID study for each one.

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